

Dylan Berger and Nicholas Graham
November 4, 2025

Design Procedure

In this lab we were tasked to implement a counting bit algorithm and a binary search algorithm. Key3 starts the systems, Key0 resets the systems, SW9 switches between the two systems.

Task 1:

An 8 bit input is given by SW7-0 for the bit counting algorithm. A 7 segment display displays the number of 1's in A. We used a second DFF module for each input to handle metastability. Key0 was our reset signal and Key3 was our start signal. We just analyzed the first bit, if it was 1 then increment, then shift right until out of bits.

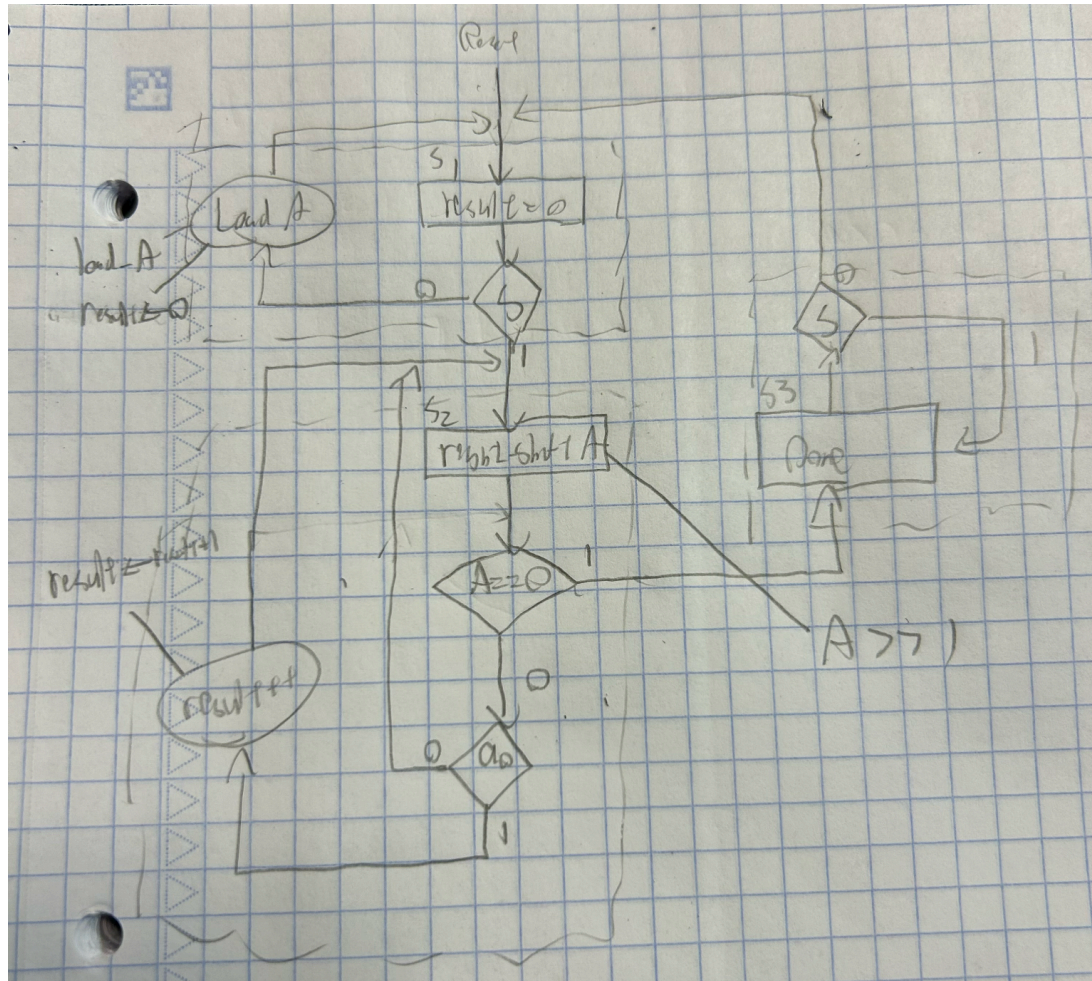


Figure 1: bit counter ASMD

Task 2:

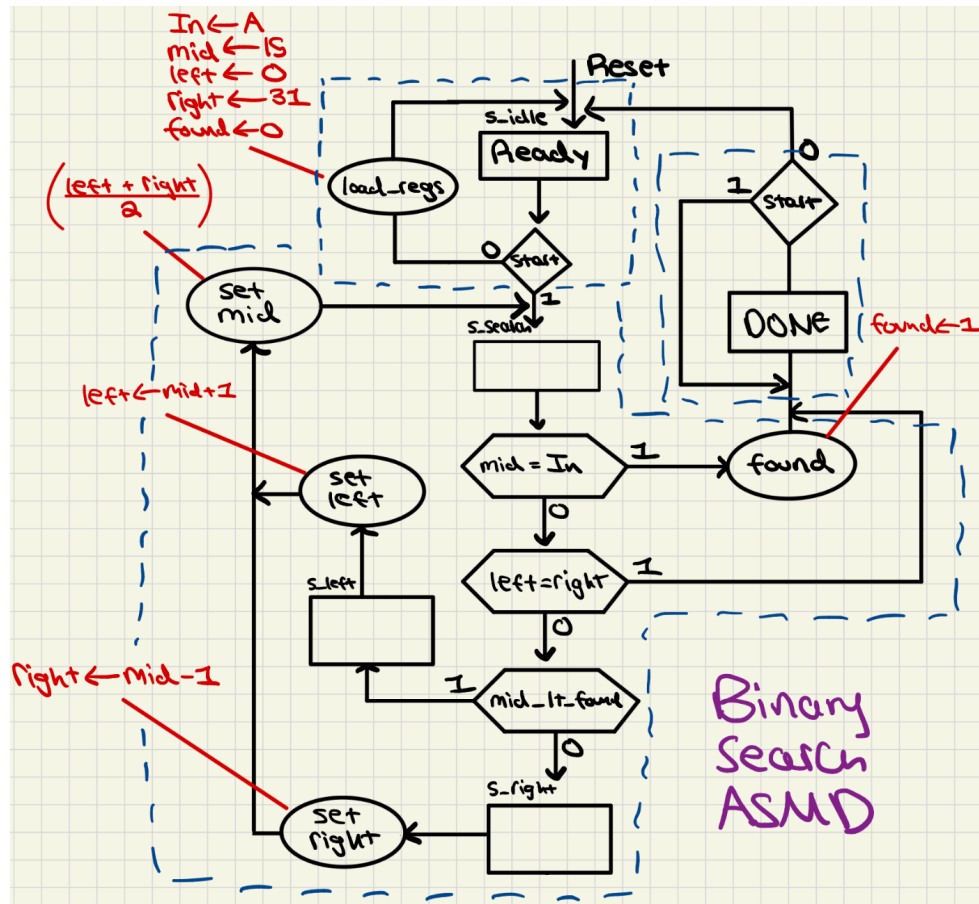


Figure 2: Binary Search ASMD

Make 32x8 RAM to register values, the RAM has a sorted collection of values. Input A, using the same implementation as Task 1. The system will then take the A value and use binary search to find where in the RAM the value is. When its finished, it will display the address on HEX1-HEX0 and it will be “done” when LEDR9 is on. LEDR0 determines if it is found or not as input A may not be in the register.

Results:

Task 1:

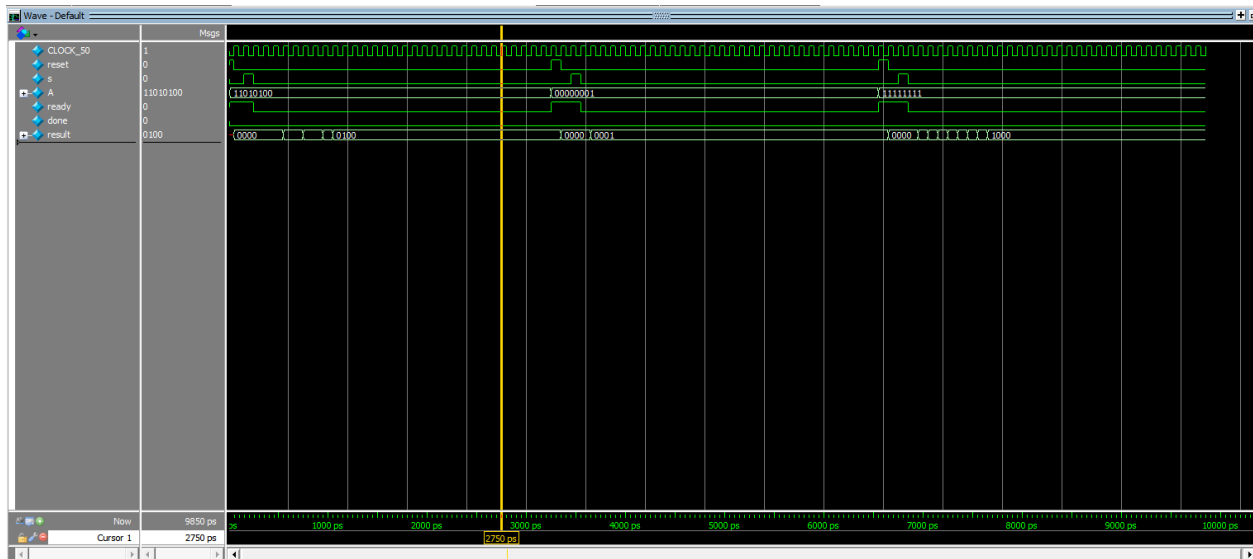


Figure 3: Task 1 simulation

We tested the bit counter on three examples: 11010100, 00000001, 11111111. The algorithm correctly outputs 0100 (4), 0001 (1), and 1000 (8) respectively for the number of 1's in the input.

Task 2:

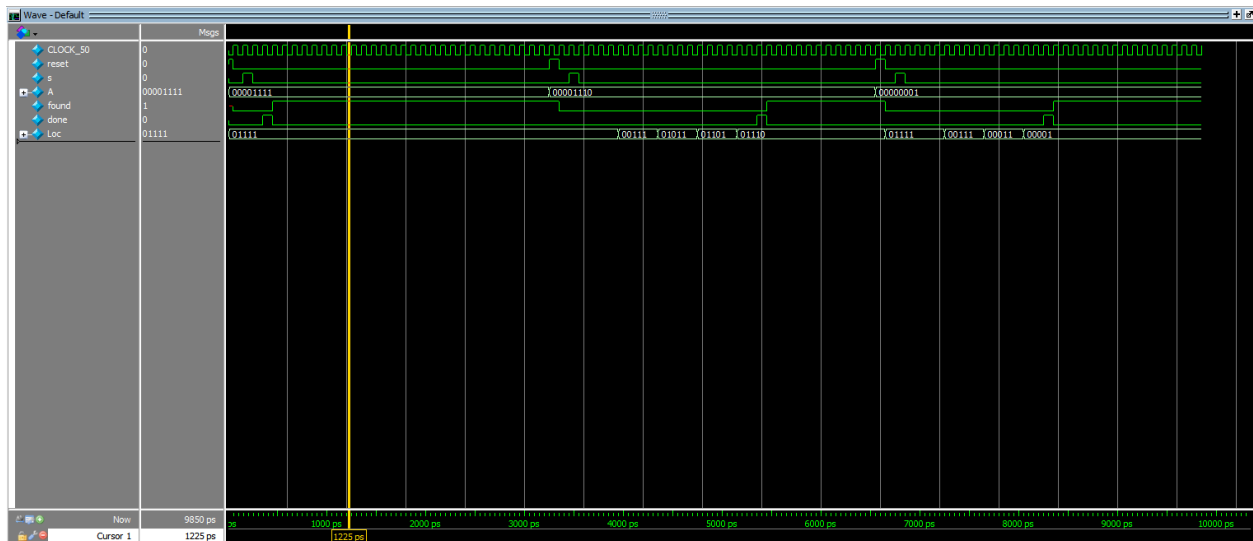


Figure 4: Task 2 simulation

We tested the binary search algorithm on three examples: 00001111, 00001110, and 00000001. Their value is the same as their address in the RAM module. As intended, the algorithm is able to find each of these and output the address for which they are located (01111,

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01110, and 00001 respectively). You can see through Loc, that the algorithm searches through the addresses as a binary search algorithm would - finding the mid of the current range of addresses.

<table> <tr><td>Flow Status</td><td>Successful - Tue Nov 04 20:57:46 2025</td></tr> <tr><td>Quartus Prime Version</td><td>17.0.0 Build 595 04/25/2017 SJ Lite Edition</td></tr> <tr><td>Revision Name</td><td>DE1_SoC</td></tr> <tr><td>Top-level Entity Name</td><td>task1_top</td></tr> <tr><td>Family</td><td>Cyclone V</td></tr> <tr><td>Device</td><td>5CSEMA5F31C6</td></tr> <tr><td>Timing Models</td><td>Final</td></tr> <tr><td>Logic utilization (in ALMs)</td><td>N/A</td></tr> <tr><td>Total registers</td><td>14</td></tr> <tr><td>Total pins</td><td>32</td></tr> <tr><td>Total virtual pins</td><td>0</td></tr> <tr><td>Total block memory bits</td><td>0</td></tr> <tr><td>Total DSP Blocks</td><td>0</td></tr> <tr><td>Total HSSI RX PCSs</td><td>0</td></tr> <tr><td>Total HSSI PMA RX Deserializers</td><td>0</td></tr> <tr><td>Total HSSI TX PCSs</td><td>0</td></tr> <tr><td>Total HSSI PMA TX Serializers</td><td>0</td></tr> <tr><td>Total PLLs</td><td>0</td></tr> <tr><td>Total DLLs</td><td>0</td></tr> </table>	Flow Status	Successful - Tue Nov 04 20:57:46 2025	Quartus Prime Version	17.0.0 Build 595 04/25/2017 SJ Lite Edition	Revision Name	DE1_SoC	Top-level Entity Name	task1_top	Family	Cyclone V	Device	5CSEMA5F31C6	Timing Models	Final	Logic utilization (in ALMs)	N/A	Total registers	14	Total pins	32	Total virtual pins	0	Total block memory bits	0	Total DSP Blocks	0	Total HSSI RX PCSs	0	Total HSSI PMA RX Deserializers	0	Total HSSI TX PCSs	0	Total HSSI PMA TX Serializers	0	Total PLLs	0	Total DLLs	0	<table> <tr><th colspan="2">Flow Summary</th></tr> <tr><td colspan="2"><<Filter>></td></tr> <tr><td>Flow Status</td><td>Successful - Tue Nov 04 20:59:03 2025</td></tr> <tr><td>Quartus Prime Version</td><td>17.0.0 Build 595 04/25/2017 SJ Lite Edition</td></tr> <tr><td>Revision Name</td><td>DE1_SoC</td></tr> <tr><td>Top-level Entity Name</td><td>task2_top</td></tr> <tr><td>Family</td><td>Cyclone V</td></tr> <tr><td>Device</td><td>5CSEMA5F31C6</td></tr> <tr><td>Timing Models</td><td>Final</td></tr> <tr><td>Logic utilization (in ALMs)</td><td>N/A</td></tr> <tr><td>Total registers</td><td>28</td></tr> <tr><td>Total pins</td><td>44</td></tr> <tr><td>Total virtual pins</td><td>0</td></tr> <tr><td>Total block memory bits</td><td>128</td></tr> <tr><td>Total DSP Blocks</td><td>0</td></tr> <tr><td>Total HSSI RX PCSs</td><td>0</td></tr> <tr><td>Total HSSI PMA RX Deserializers</td><td>0</td></tr> <tr><td>Total HSSI TX PCSs</td><td>0</td></tr> <tr><td>Total HSSI PMA TX Serializers</td><td>0</td></tr> <tr><td>Total PLLs</td><td>0</td></tr> <tr><td>Total DLLs</td><td>0</td></tr> </table>	Flow Summary		<<Filter>>		Flow Status	Successful - Tue Nov 04 20:59:03 2025	Quartus Prime Version	17.0.0 Build 595 04/25/2017 SJ Lite Edition	Revision Name	DE1_SoC	Top-level Entity Name	task2_top	Family	Cyclone V	Device	5CSEMA5F31C6	Timing Models	Final	Logic utilization (in ALMs)	N/A	Total registers	28	Total pins	44	Total virtual pins	0	Total block memory bits	128	Total DSP Blocks	0	Total HSSI RX PCSs	0	Total HSSI PMA RX Deserializers	0	Total HSSI TX PCSs	0	Total HSSI PMA TX Serializers	0	Total PLLs	0	Total DLLs	0
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Family	Cyclone V
Device	5CSEMA5F31C6
Timing Models	Final
Logic utilization (in ALMs)	N/A
Total registers	42
Total pins	44
Total virtual pins	0
Total block memory bits	128
Total DSP Blocks	0
Total HSSI RX PCSs	0
Total HSSI PMA RX Deserializers	0
Total HSSI TX PCSs	0
Total HSSI PMA TX Serializers	0
Total PLLs	0
Total DLLs	0

Figure 7: Combined Task 1 and Task 2 Flow Summary

Experience Report

Lab was quick and rewarding. Submitted late due to backlog from other assignments and classes, but flowed very smoothly.

This lab took us approximately 5 hours, broken down as follows:

- Reading – 20 minutes
- Planning – 10 minutes
- Design – 30 minutes
- Coding – 2 hours
- Testing/debugging – 2 hours